# Project Proposal: Smart Email Triage Assistant for Small Business Owners

## Problem Statement & Motivation

### What Real-World Business Problem Are You Addressing?

My project addresses the business problem of inefficient email triage and response management for small business owners who need a more efficient way to manage their inbox.

Small business owners and solo entrepreneurs are responsible for managing nearly every aspect of their operations, including customer communication. They receive a steady stream of emails such as customer inquiries, scheduling requests, invoices, vendor communication, marketing messages, and spam. Manually reading, sorting, and responding to these emails consumes valuable time and can lead to delayed responses, missed urgent messages, or inconsistent communication.

While modern email platforms offer basic tools such as folders, filters, and keyword-based rules, these approaches are limited. They do not reliably capture email intent, urgency, or sentiment, and they require ongoing manual configuration. As inbox volume increases, static rules become ineffective, reducing productivity and negatively impacting customer satisfaction.

### Who Are the Target Users?

The target users are **small business owners or solo entrepreneurs** who do not have a dedicated administrative assistant and are responsible for managing their own email inbox.

### Why Is an Agentic AI Solution Appropriate?

An agentic AI solution is appropriate because email management involves more than simple automation. Effective triage requires interpreting intent, assessing urgency, generating appropriate responses, and reviewing outputs before action is taken. These tasks involve reasoning, judgment, and iteration.

A static workflow defined by a human or simple automation rules cannot adapt to ambiguous or multi-intent emails. In contrast, an agentic AI system can autonomously analyze incoming emails, make decisions about priority and response strategy, and revise outputs through internal critique. A multi-agent approach mirrors real-world roles such as sorting, responding, and quality control, making it both realistic and effective

## Use Cases & Scenarios

#### Scenario 1: Daily Inbox Organization

A small business owner wants to quickly understand which emails require immediate attention.

- The system receives a batch of incoming emails.
- One agent categorizes emails by type (customer inquiry, billing, scheduling, low priority).
- Another agent evaluates urgency and flags high-priority messages.
- The system presents a summarized and prioritized inbox view to the user.

**Scenario 2: Drafting Customer Responses**

A customer sends an email requesting information about pricing or availability.

- The system extracts key details from the email.
- A response-drafting agent generates a professional reply.
- A reviewing agent checks the response for tone, clarity, and relevance.
- The final draft is shown to the user for approval or editing.

**Scenario 3: Identifying Sensitive or Urgent Emails**

A customer email expresses frustration or dissatisfaction.

- The system detects negative sentiment or urgency.
- The email is flagged as high priority.
- A response draft emphasizes empathy and timely resolution, helping the user respond appropriately.

# Initial System Design

## Preliminary List of Agents and Their Roles

- **Email Classification Agent**: Categorizes emails by topic and intent.
- **Priority Assessment Agent**: Determines urgency and importance based on content and sentiment.
- **Response Drafting Agent**: Generates response drafts for actionable emails.
- **Reviewer / Critic Agent**: Evaluates classifications and drafts, identifies errors, and provides feedback to other agents.

These agents operate in a coordinated, multi-agent workflow where outputs are reviewed and refined before being presented to the user.

## Expected Tools, APIs, or Data Sources

- Large Language Model (LLM) API for reasoning, classification, and text generation
- Local dataset or JSON files containing simulated email messages
- File-based storage for categorized email metadata
- Streamlit front-end for user interaction and visualization

Live email integrations are excluded to maintain feasibility within the project scope.

## Feasibility & Risks

**Potential Challenges**

- **Hallucination**: The system may misinterpret email intent or generate inaccurate responses.
- **Misclassification**: Important emails may be incorrectly labeled as low priority.
- **Tone errors**: Generated responses may not fully align with professional expectations.
- **Context limitations**: Long email threads may exceed context limits.

### Constraints

- API rate limits and token usage
- Limited computation resources
- Privacy considerations when handling email content
- Project timeline constrained to the course duration

### Mitigation Strategies

- Use structured prompts and clearly defined agent responsibilities
- Include a Reviewer Agent to validate outputs
- Limit context to the most relevant email content
- Require user approval before any response is finalized

## Timeline for Project Execution

The project will be completed within the class duration:

- **Weeks 1–2**: Finalize scope, agent design, and prepare email dataset
- **Weeks 3–4**: Implement core agents and tool interactions
- **Week 5**: Develop user-facing interface
- **Week 6**: Integrate multi-agent feedback loop
- **Final Week**: Testing, evaluation, and final report preparation